

COL774

Machin Learning
Aug 12, 2025Derivation
of $f'(0)$ Make-up class
Today Aug 12
6pm-7p
in lieu of
class on T Aug 14

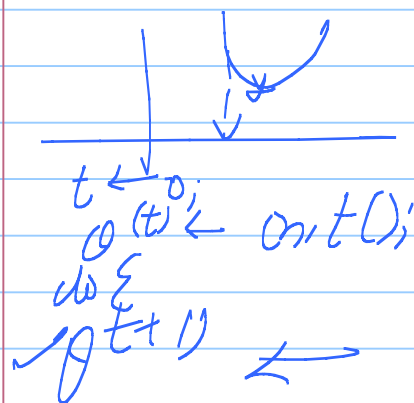
last (loss =

$$f(0) = (0 - 3)^2 + 1 \rightarrow 2(0 - 3)$$

$$\text{w.r.t } \frac{df(0)}{d0} = 0$$

in higher dim.

$$\nabla f(0) = 0$$


 $t \leftarrow t+1$
 3 while (converged)

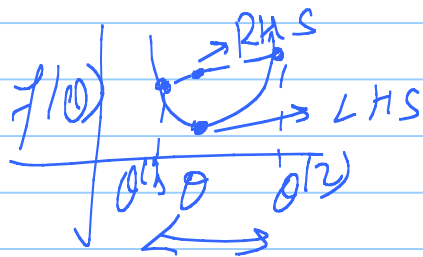
 $\hookrightarrow$ learning rate

1. When does it converge to global minima?

2. How do we detect convergence?

3. Convergence properties of this algorithm (gradient descent)

Convex function



f is convex iff $0 \leq \alpha \leq 1$

$\exists \theta^{(1)}, \theta^{(2)} \in \text{Dom}(f)$

$$\rightarrow f(\alpha \theta^{(1)} + (1-\alpha) \theta^{(2)}) \leq \alpha f(\theta^{(1)}) + (1-\alpha) f(\theta^{(2)}) \quad \text{--- Defn 1}$$

$\Rightarrow f$ is doubly differentiable

f is convex iff $\frac{\partial^2 f(\theta)}{\partial \theta^2} \geq 0 \quad \text{--- (Defn 2)}$

can also define strictly convex or strictly concave

Convex set:-

S:-

$\theta^{(1)}, \theta^{(2)} \in S$

$\alpha \theta^{(1)} + (1-\alpha) \theta^{(2)} \in S$

$0 \leq \alpha \leq 1$

↳ $0 \in \mathbb{R}^n$ Defn. is still valid $f: \mathbb{R}^n \rightarrow \mathbb{R}$

Assume ^{Symmetric} matrix Hessian: - (Matrix of second order partial derivatives)

↳ $H \equiv \begin{bmatrix} \frac{\partial^2 f(0)}{\partial \theta_1 \partial \theta_1} & \dots & \frac{\partial^2 f(0)}{\partial \theta_1 \partial \theta_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f(0)}{\partial \theta_n \partial \theta_1} & \dots & \frac{\partial^2 f(0)}{\partial \theta_n \partial \theta_n} \end{bmatrix}$ $\nabla^2 f(0) = \begin{pmatrix} \frac{\partial f(0)}{\partial \theta_1} \\ \vdots \\ \frac{\partial f(0)}{\partial \theta_n} \end{pmatrix}$

↳ $\mathbb{R}^{n \times n}$ $\downarrow \mathbb{R}^n$

Replaces $\frac{d^2 f(0)}{d\theta^2}$ in higher dimension

Given a ^{Symmetric} matrix $A \in \mathbb{R}^{n \times n}$, A is +ve semi-definite if $\forall Z \in \mathbb{R}^n \rightarrow Z^T A Z \geq 0$

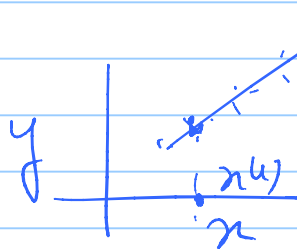
Defn $\begin{cases} \text{negative semi-definite} \rightarrow \text{if } z^T A z \leq 0 \quad \forall z \in \mathbb{R}^n \\ \text{negative definite} \rightarrow z^T A z < 0 \quad \forall z \in \mathbb{R}^n \end{cases}$

$\rightarrow A$ is +ve definite $z^T A z > 0$

$f: \mathbb{R}^n \rightarrow \mathbb{R}$ doubly differentiable then f is convex iff H (Hessian Matrix) is positive semi-definite $\} \& \text{ other way round for concave (i.e. -ve semi-definite)}$

$\rightarrow$ (Defn 2)

$\Rightarrow$ linear regression



$$h_0(n) = \theta^T x$$

$$h_0(n) = \theta^T x \quad \theta \in \mathbb{R}^{n+1}$$

$$h_0: \mathbb{R} \rightarrow \mathbb{R} \quad x \in \mathbb{R}^{n+1} \quad n_0 = 1$$

$$\theta = \begin{bmatrix} \theta_0 \\ \vdots \\ \theta_n \end{bmatrix}$$

$$x = \begin{bmatrix} x_0=1 \\ \vdots \\ x_n \end{bmatrix}$$

$\mathbb{R}^n \rightarrow \mathbb{R}$

$$h_{\theta}(x) = \theta^T x$$

Loss function (MSE) Mean Squared Error

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m \left[y^{(i)} - \underbrace{h_{\theta}(x^{(i)})}_{\text{Output}} \right]^2$$

argmin _{θ} $J(\theta)$

$\mathbb{R}^n \times \mathbb{R}$

For linear Regression, $h_{\theta}(x) = \theta^T x$

$$\Rightarrow J(\theta) = \frac{1}{2m} \sum_{i=1}^m \left[y^{(i)} - \theta^T x^{(i)} \right]^2$$

$$V_0 J(0) = \left[- \frac{1}{2g} \frac{\partial J(0)}{\partial g} \right]$$

$$\partial \left\{ 0 \dots n+1 \right\} \frac{\partial J(0)}{\partial g} = \frac{1}{2m} \sum_{i=1}^m \frac{\partial}{\partial (\theta^T x^{(i)})} \underbrace{\left(y^{(i)} - \theta^T x^{(i)} \right)}_{(y^{(i)} - \theta^T x^{(i)})^2} \cdot \underbrace{\frac{\partial \theta^T x^{(i)}}{\partial \theta_j}}_{\frac{\partial \theta^T x^{(i)}}{\partial \theta_j}}$$

$$\theta^T x^{(i)} = \sum_{j=0}^{n+1} \theta_j x_j^{(i)}$$

$$\frac{\partial}{\partial (y^{(i)} - \theta^T x^{(i)})} \underbrace{(y^{(i)} - \theta^T x^{(i)})^2}_{(y^{(i)} - \theta^T x^{(i)})^2} \cdot \underbrace{\frac{\partial (y^{(i)} - \theta^T x^{(i)})}{\partial \theta_j}}_{\frac{\partial \theta^T x^{(i)}}{\partial \theta_j}}$$

$$\frac{1}{2m} \sum_{i=1}^m \frac{\partial}{\partial (y^{(i)} - \theta^T x^{(i)})} (-1) x_j^{(i)}$$

$$\Rightarrow \frac{\partial J(\theta)}{\partial \theta_j} = \frac{1}{m} \sum_{i=1}^m (y^{(i)} - \theta^T x^{(i)}) (-1) x_j^{(i)}$$

$$\boxed{\nabla_{\theta} J(\theta) = \frac{1}{m} \sum_{i=1}^m (y^{(i)} - \theta^T x^{(i)}) (-1) x^{(i)}} \quad \text{--- (1)}$$

↳ Gradient descent update:

$$\theta^{(t+1)} \leftarrow \theta^{(t)} - \eta \cdot \nabla_{\theta} J(\theta) / \theta^{(t)}$$